

S1 File Supporting Information: Sensitivity Attributes Definitions

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See Morrison et al. (2015) for more details.

Morrison W, Nelson M, Howard J, Teeters E, Hare JA, Griffis R. et al. Methodology for assessing the vulnerability of fish stocks to a changing climate. NOAA Technical Memorandum. 2015; NMFS-OSF-3: 1-48.

<http://www.st.nmfs.noaa.gov/Assets/ecosystems/climate/documents/TM%20OSF3.pdf>

Background

The goal of this project is to provide regional fisheries managers with a practical tool to efficiently assess the vulnerability of fish stocks to climate change. *Vulnerability* is defined here as the extent to which the abundance and productivity of a stock could be impacted by climate change. The potential for a change in distribution and the directional effect (positive or negative) of a changing climate are also assessed. This project considers the overall vulnerability of fish stocks to climate change to be a function of two main components: **exposure** and **sensitivity**.

Exposure is a measure of the predicted environmental change that the stock may experience within its range. It is the overlap between the stock's distribution and the magnitude of and spatial distribution of the expected environmental change. The factors accounted for in exposure may include increases in temperature, changes to freshwater input, rise in sea level, changes to ocean circulation, etc.

The *sensitivity* component is composed of biological attributes which are believed to be indicative of the response that a stock will have to potential changes in climate. Sensitivity includes the attributes that describe the stock's resilience (the ability of a stock to survive and recover from a perturbation) and its adaptive capacity (the ability of a stock to adapt, reduce or mitigate the consequences through evolutionary changes and plastic ecological responses) (Williams et al. 2008).

This document provides definitions, justifications, links to climate change and scoring bins for each of the sensitivity attributes. This vulnerability assessment can be completed at either the species level or the stock level, depending on the goals of the assessment. We have used the term "stock" throughout this document; however, where appropriate, it can be replaced by "species" if that is the level of analysis being implemented.

This methodology leans heavily on expert opinions. Experts should use their expert knowledge when using and interpreting these attributes and attribute bins. For example, experts may encounter a situation where the scoring bins suggest a specific attribute score, but their expert knowledge of the species or data makes them think the score should be higher, lower, or more uncertain. We are counting on the experts to make these calls. However, experts should be prepared to provide justification for their scores.

Data quality will also be scored by experts. Please refer to Table 1 on the following page for a scoring rubric. Experts should use their expert knowledge when evaluating data quality for each sensitivity attribute.

Data Quality

Table 1. Data quality scoring guidelines.

Data Quality Score	Description
3	Adequate Data. The score is based on data which have been observed, modeled or empirically measured for the species in question and comes from a reputable source.
2	Limited Data. The score is based on data which has a higher degree of uncertainty. The data used to score the attribute may be based on related or similar species, come from outside the study area, or the reliability of the source may be limited.
1	Expert Judgment. The attribute score reflects the expert judgment of the reviewer and is based on their general knowledge of the species, or other related species, and their relative role in the ecosystem.
0	No Data. No information to base an attribute score on. Very little is known about the species or related species and there is no basis for forming an expert opinion.

Sensitivity Attributes

Habitat Specificity

Goal: To determine the breadth of habitat used by a stock and estimate the ability of individuals to shift habitat use under changing conditions by taking into consideration if the stock is a habitat generalist or specialist, and the type and sensitivity of the habitat.

Relationship to climate change: Stocks that are reliant on specific habitat types may be more vulnerable to climate change because they are dependent on not only their own response to climate change, but also the impact on their habitat (EPA 2009). Note: the type (biotic vs. abiotic) and distribution of these habitats should be considered for this attribute.

Background: Fish stocks rely on biophysical features (i.e., habitat) for shelter, foraging, and spawning throughout various life stages. Species that rely on specific physical and biological features are more likely to be sensitive to climate change, especially if the features are vulnerable to climate-driven changes (Morrison et al. 2015). Reliance on different types of features is expected to result in different levels of sensitivity.

For the purpose of this assessment, we consider three types of habitat: physical habitat expected to be resilient to changing climate conditions, physical habitat expected to be vulnerable to changing climate conditions, and biogenic habitat, which is expected to be vulnerable to climate change.

Physical features such as depth, bathymetry, and rocky shorelines are expected to be resilient to climate change and therefore would result in lower sensitivity for those species that rely on those types of habitat. Other physical features that are more vulnerable to climate changes (e.g., sea ice, beach topography, fronts, eddies, upwelling) will produce greater impacts to species that rely on those types of features. We consider the physical and chemical characteristics of the water column as habitat features, which are more dynamic than persistent geologic features.

Biogenic habitat – habitat created by or consisting of organisms or organism remains – may undergo the greatest changes from a changing climate, as both the ecosystem engineers and underlying physical conditions may be impacted by changing conditions (e.g., Nelson 2009, Doney et al. 2012, Harley et al. 2012). Examples of biogenic habitat include kelp forests, mangroves, salt marshes, coral reefs, and seagrass beds (Teck et al. 2010, Okey et al. 2015). Thus, species that depend on biogenic habitats are likely more vulnerable to climate change. While the presence of suitable prey plays a key role in defining a species' habitat, we consider the prey and diet specificity of the species in a separate attribute.

Similar to the prey/diet specificity attribute, we encourage future iterations of this assessment to interface with vulnerability assessments that score the vulnerability of habitat to climate change.

All aspects of a stock's life history within and outside of US waters should be considered when scoring this attribute.

How to use expert opinion: This attribute will be scored using a combination of the three criteria described above: abundance, distribution, and phenology. Please weigh the goals listed above equally. .

It is understood that these criteria are not dichotomous but are a continuum. If you think that a stock fits in multiple scoring bins, weight your 5 tallies between the appropriate bins. Using your expert opinion, account for any lifespan or ontogenetic shifts in diet; however, limit your response to the juvenile and adult life stages as larvae are considered under the attribute "early life history survival."

Habitat Specificity Bins:

1. **Low:** Stock exclusively utilizes physical features resilient to climate conditions, and is not reliant on biogenic habitat.
2. **Moderate:** Stock utilizes a variety of features or prefers a particular habitat, but is not reliant on physical features vulnerable to climate conditions and/or biogenic habitat for specific life stages. The stock can survive in other habitats (with possible impacts to their fitness).
3. **High:** Stock relies on biogenic habitat or physical features vulnerable to climate conditions for one life stage or event.
4. **Very High:** Stock relies on biogenic habitat or physical features vulnerable to climate conditions for multiple life stages or events, or for any one particularly critical life stage or event.

Prey Specificity

Goal: To determine, on a relative scale, if the stock is a prey generalist or a prey specialist.

Relationship to climate change: Understanding how reliant a stock is on specific prey species could predict its ability to persist as the climate changes. Generalists (who feed across a wide spectrum of prey types) should have a better chance to persist in response to a changing environment. Alternatively, specialists (who have specific prey requirements) are likely to be more vulnerable to climate change because their persistence is dependent on not only their own response to climate change, but also the response of their prey.

Background: Climate change impacts extend beyond the stock in question to include species within its food web (e.g., prey, predators and competitors).

How to use expert opinion: The scoring bins below estimate the stocks' relative distribution along a continuum that runs between prey specialists and prey generalists. Using your expert opinion, account for any lifespan or ontogenetic shifts in diet; however, limit your response to the juvenile and adult life stages as larvae are considered under the attribute "early life history survival and settlement requirements." For this attribute, prey type refers to groups of similar species; copepods, krill, forage fish, etc., for example, are each categorized as a prey type.

Prey Specificity Bins:

1. **Low: The stock eats a large variety of prey.** The stock can eat a variety of prey types depending on what is available and can easily switch among prey types. There are no known predator-prey circumstances affecting this stock that might affect accessibility to prey or overall fitness. Include detritivores, herbivores, and omnivores in this bin.
2. **Moderate: The stock eats a limited number of prey types.** The stock can feed on a wide variety of prey, but are restricted to a limited number of prey types (e.g., copepods, krill, forage fish, etc). While they may be restricted to specific prey categories, they can easily switch among them. There are no known predator-prey circumstances affecting this stock that might affect accessibility to prey or overall fitness.
3. **High: The stock is partial to a single prey type.** The stock's diet is composed of one main prey type. The stock is able to switch to a different prey type if the preferred food is unavailable, but this may negatively impact fitness. There are known predator-prey circumstances affecting this stock documented in the literature.
4. **Very High: The stock is a specialist.** The stock is dependent on one prey type and is unable to switch to alternatives if the preferred prey is unavailable. There are known predator-prey circumstances affecting this stock documented in the literature.

Sensitivity to Ocean Acidification (OA)

Goal: To estimate a stock's sensitivity to ocean acidification based on its relationship with "sensitive taxa."

Relationship to climate change: Impacts of OA on marine organisms can be highly variable, with considerable variability between taxa and species (Kroeker et al. 2013). Therefore, we are estimating the impact of OA by examining the dependence of the stock on sensitive taxa. For example, current research shows a consistent negative impact of OA on mollusks and corals, so species in either of these classes or dependent on species in these classes should be considered more sensitive to changes in ocean pH. We expect the volume of research into ocean acidification to increase in the near future, so this attribute will be updated as new information becomes available.

Background: Ocean acidification is often called "the other carbon dioxide problem," and is the term given to the chemical changes in the ocean as a result of carbon dioxide emissions (Wicks and Roberts 2012). While initial research suggested that the majority of species that have calcium carbonate or chitin shells or those that lay down calcium carbonate skeletons (corals) will be negatively impacted by ocean acidification (Arnold et al. 2009; Hoegh-Guldberg et al. 2007; Honisch et al. 2012; Kawaguchi et al. 2011; Orr et al. 2005), recent studies have highlighted a high variability in response between different shelled organisms and suggest that not all shelled species will be impacted to the same degree and not all impacts will be negative. (i.e., Ries et al. 2009, Kroeker et al. 2013). For example, Kroeker et al. (2013) in a meta-analysis of 228 studies found significant and consistent negative impacts of OA on the larval stages of mollusks and corals (see Figure 4 from Kroeker et al. below). In contrast, high variability in the responses of crustaceans suggests impacts may be species specific within this group, with brachyuran crustaceans showing a higher resistance (Kroeker et al. 2013).

The direct effect of ocean acidification on finfish is not well understood. Recent research suggests impacts on finfish stocks will be most prevalent at the egg and early larval stages (Baumann et al. 2011; Franke and Clemmenssen 2011; Frommel et al. 2011), but juvenile and adult olfaction may also be affected (Mundy et al. 2009). Where available, this CVA reviews existing literature on the effects of ocean acidification on finfish throughout all life stages.

How to use expert opinion: Use the results presented in Figure 4 from Kroeker et al. 2013 (or other relevant information) to bin species. When scoring, base your score on the most sensitive life stage, if appropriate. In cases where research has shown that the effects of OA may be positive or mitigated by biological processes (e.g. reduced OA by plant absorption of CO₂), use your expert judgment to inform the score.

Sensitivity to Ocean Acidification Bins:

1. **Low: Stock not reliant on sensitive taxa.** The stock does not utilize sensitive taxa for food or habitat. Species expected to respond positively to ocean acidification should be scored as low.
2. **Moderate: Stock is somewhat reliant on sensitive taxa.** The stock utilizes sensitive taxa as either food or habitat. This can include omnivores and species that prefer coral habitats but can utilize any rigid structure.
3. **High: Stock is reliant on sensitive taxa.** The stock is highly dependent on sensitive taxa for either food or habitat (i.e., cannot switch to a non-sensitive alternative).
4. **Very High: Stock is a sensitive taxa.** The stock is a sensitive taxa (such as corals or mollusks) that have been shown to have a consistent negative impact of OA on survival, growth or abundance.

Taxa	Response	Mean Effect	
 Calcifying algae	Survival		Not tested or too few studies
	Calcification		Enhanced <25%
	Growth		95% CI overlaps 0
	Photosynthesis	-28%	Reduced <25%
	Abundance	-80%	Reduced >25%
 Corals	Survival		Not tested or too few studies
	Calcification	-32%	Reduced >25%
	Growth		95% CI overlaps 0
	Photosynthesis		95% CI overlaps 0
	Abundance	-47%	Reduced >25%
 Coccolithophores	Survival		Not tested or too few studies
	Calcification	-23%	Reduced <25%
	Growth		95% CI overlaps 0
	Photosynthesis		95% CI overlaps 0
	Abundance		95% CI overlaps 0
 Mollusks	Survival	-34%	Reduced >25%
	Calcification	-40%	Reduced >25%
	Growth	-17%	Reduced <25%
	Development	-25%	Reduced <25%
	Abundance		95% CI overlaps 0
 Echinoderms	Survival		95% CI overlaps 0
	Calcification		95% CI overlaps 0
	Growth	-10%	Reduced <25%
	Development	-11%	Reduced <25%
	Abundance		Not tested or too few studies
 Crustaceans	Survival		95% CI overlaps 0
	Calcification		95% CI overlaps 0
	Growth		95% CI overlaps 0
	Development		95% CI overlaps 0
	Abundance		95% CI overlaps 0
 Fish	Survival		Not tested or too few studies
	Calcification		Not tested or too few studies
	Growth		95% CI overlaps 0
	Development		Not tested or too few studies
	Abundance		Not tested or too few studies
 Fleshy algae	Survival		Not tested or too few studies
	Calcification		Not tested or too few studies
	Growth	+22%	Enhanced <25%
	Photosynthesis		95% CI overlaps 0
	Abundance		95% CI overlaps 0
 Seagrasses	Survival		Not tested or too few studies
	Calcification		Not tested or too few studies
	Growth		95% CI overlaps 0
	Photosynthesis		95% CI overlaps 0
	Abundance		Not tested or too few studies
 Diatoms	Survival		Not tested or too few studies
	Calcification		Not tested or too few studies
	Growth	+17%	Enhanced <25%
	Photosynthesis	+12%	Enhanced <25%
	Abundance		95% CI overlaps 0

Reproductive Strategy Sensitivity

Goal: To determine how sensitive the stock's reproductive strategy is and how dependent reproductive success is on specific environmental conditions.

Relationship to climate change: Species that have complex reproductive strategies (that require a series of events or special conditions) are more likely to be sensitive to changes in the environment.

Background: There is great diversity in reproductive strategies in marine fishes. The more complex the reproductive strategy, the more precise the conditions may need to be, and thus the more vulnerable the stock may be to environmental change. For our purposes, complexity in reproductive strategy is defined as reproductive behaviors, characteristics or cues that create specific requirements that must be met in order for reproduction to be successful.

How to use expert opinion: A list of common reproductive characteristics that may affect the reproductive capacity of a stock in a changing climate is provided below. To score, determine if any of these examples apply to the stock. Note: this is not intended to be an exhaustive list. If other characteristics exist that may affect a stock's reproduction capacity in a changing climate, incorporate that information and adjust your score appropriately.

Example reproductive characteristics that range in sensitivity include but are not limited to:

- The stock has known temperature effects on reproduction. Examples include temperature-dependent sex changes, and temperature cues that impact spawning, gonad development, etc.
- The stock uses large spawning aggregations and displays site fidelity to specific spawning grounds. Large spawning aggregations can contribute to a high sensitivity because a large number of individuals must get to the spawning area simultaneously (i.e., migration or cues to migrate may be impeded by a change in the environment), the spawning area has to retain the environmental conditions that made it successful in the past, and the reproductive success for that year is dependent on the conditions present at one time period.
- The stock experiences decreased recruitment at low stock sizes due to depensation/allee effects. If this is not known, does the stock share life history characteristics that would predict strong allee effects (e.g., at low densities, urchins can experience decreased fertilization and thus reduced recruitment)?
- The reproductive success of the stock requires the use of vulnerable habitats (freshwater, estuaries, mangroves, salt marshes, coral reefs) for spawning or rearing young. Vulnerable habitats are likely to experience larger climate change impacts (such as changes in salinity, dissolved oxygen, pollution, sedimentation, or water depth), and stocks that require these habitats for successful reproduction will likely be impacted.

Reproductive Strategy Sensitivity Bins:

1. **Low: Simple reproductive strategy.** The stock contains no more than one characteristic

that suggests complexity in reproductive strategy.

2. **Moderate: Slight complexity.** The stock has two characteristics that suggest complexity in reproductive strategy.
3. **High: Complex reproductive strategy.** The stock has three characteristics that suggest complexity in reproductive strategy.
4. **Very High: Very complex reproductive strategy.** The stock has four or more characteristics that suggest complexity in reproductive strategy.

Sensitivity to Temperature

Goal: To use information regarding temperature of occurrence or the distribution of the species as a proxy for its sensitivity to temperature. Note that this attribute uses species (vs. stock) distributions as they better predict thermal requirements.

Relationship to climate change: Species that experience a wide range of temperature regimes are more likely to persist in a warming ocean (Chin et al. 2010).

Background: The distribution of a species within or across provinces provides an estimate of its temperature requirements. Spalding et al. (2007) divides coastal waters of the world into 62 provinces and 232 ecoregions. Even though Spalding's provinces are not specifically based on temperature (they also consider upwelling, currents, salinity, nutrients, etc.), they can be used to delineate areas with similar thermal conditions.

In addition, a species' distribution in the water column and seasonal movements can indicate its sensitivity to temperature. Species that make large diurnal migrations across the thermocline have lower sensitivities to changing temperatures than species that have limited depth distributions. Additionally, species that make large seasonal migrations and track seasonally changing water temperatures may have more sensitivity to temperature than indicated by range alone.

How to use expert opinion: Use known temperature requirements to score this attribute when available. When temperature information is not known, use the species distribution, along with Spalding et al. (2007) to determine if a species is found across >1 province. Also use knowledge of seasonal and diurnal movements to adjust the tallies. Keep in mind that you can adjust your tallies depending on the distribution of the species relative to the area of interest (i.e. if the area of interest is at the edge of the distribution of the species, consider if the species is expected to move out of or expand into the area of interest). Spalding et al. (2007) only characterize coastal environments; therefore, use your expert opinion for open ocean species. If information about temperature requirements or depth distributions is available, use this to modify your response. For example, if a species is found across 2 provinces, but it has a limited depth distribution, the expert could distribute the 5 tallies between bins 2 and 3. If a species' sensitivity changes with ontogeny, consider the most limited stage when determining the most appropriate bin(s).

Temperature Sensitivity Bins:

1. **Low: Large temperature range.** Species occurs in a wide range of temperatures (>15°C), or is found across 3 or more provinces.
2. **Moderate: Moderate temperature range.** Species occurs in a moderately wide range of temperatures (10-15°C), or is found across 2 provinces.
3. **High: Somewhat limited temperature range.** Species occurs in a moderately narrow range of temperatures (5-10°C), or is found within one province but has a variable depth distribution.

4. **Very High: Very limited temperature range.** Species occurs in a narrow range of temperatures ($<5^{\circ}\text{C}$), or is found within one province and has a limited depth distribution (i.e., depth range is $<100\text{ m}$).

Specificity in Early Life History Requirements

Goal: To determine the relative importance of early life history requirements for a stock.

Relationship to climate change: In general, the early life stages (eggs and larvae) of marine fish are characterized by high mortality rates, via predation, starvation, advection, or unsuitable conditions. Small changes in the environment can lead to large changes in early life survival, which can affect recruitment and year-class strength.

Background: Close to 100 years ago, fisheries scientists recognized the importance of recruitment variability in fish populations (Hjort 1914). Since then, multiple hypotheses have been developed to explain this variability, but scientists now understand that multiple processes are important during the egg and larval stages (Houde 2008). Conditions that can lead to decreased or negligible recruitment include:

- Larvae that are dependent on specific biological conditions in the water column during their larval stage. For example, if the larvae are dependent on the presence of food at a specific point in development, different emergence of the larvae and the food (due to dependence on different cues) could result in a mismatch in availability. Alternatively, if the larvae have evolved to survive in low predator (and low food) conditions, a change in predation pressure could impact survival (Bakun 2010).
- Larvae that are dependent on specific physical conditions to survive (e.g., temporary gyres that provide food and retention, calm conditions that allow for concentration of prey, specific transport pathways to nursery habitats, etc.).
- Larvae that are dependent on a cue that could be impacted by a changing climate.

For the purpose of this assessment, early life history requirements include the environmental conditions necessary for larval survival, and encompass the eggs and pelagic larvae stages. The more specific the early life history requirements, the more precise the environmental conditions may need to be, and thus the more vulnerable the stock may be in a changing environment. Note: some fish species, namely elasmobranchs, have evolved life history traits which minimize or eliminate early life stages either by birthing well-developed young or by laying egg cases that allows embryos to fully develop before hatching. Therefore, elasmobranchs should be ranked as “Low.”

How to use expert opinion: Marine species are largely dependent on both physical and biological conditions during their larval stage. However, the specificity of these conditions varies between stocks. If no citable reference is available, the score may be based on expert opinion.

Specificity in Early Life History Requirements Bins:

1. **Low: Larval requirements are minimal.** Stock has general requirements for the larval

stage that are relatively resilient to environmental change. Elasmobranchs should be ranked as “Low.”

2. **Moderate: Larval requirements are minimal or unknown.** Stock requirements are not well understood and recruitment is relatively constant, suggesting limited environmental influence.
3. **High: Larvae have some specific requirements.** Stock requirements are not well understood, but recruitment is highly variable and appears to have a strong dependence on environmental conditions.
4. **Very High: Larvae have multiple specific requirements.** Stock has specific known biological and physical requirements for larval survival.

Stock Size/Status

Goal: To estimate stock status to clarify how much stress from fishing the stock is experiencing and to determine if the stock's resilience or adaptive capacity are compromised due to low abundance.

Relationship to climate change: It is assumed that a stock that has a large biomass is more resilient to changes in climate. Conversely, stocks with very low biomass are likely to be in a compromised ecological position and therefore may have a diminished capability to respond to climate change (Rose 2004). The genetic diversity, as well as the abundance, of a stock can impact its susceptibility. The assumption is that species with a limited genetic diversity could be more negatively impacted by climate change as their offspring would be less variable and thus less likely to have the combination of genes needed to adapt to changes in the environment. Note: stocks that are at historical high biomass levels may be an indication of a net positive effect to an environmental change.

Background: Fish stocks that are already being affected by other stressors are likely to have faster and more acute reactions to climate change. Fishing is the largest stressor currently impacting fish stocks (Jackson et al. 2001), and the magnitude of the stress can be estimated through the status of the stock. Different biological reference points are used for Atlantic HMS assessed domestically through the Southeast Data, Assessment and Review (SEDAR) process (i.e., most sharks) versus Atlantic HMS that are assessed internationally by the International Commission for the Conservation of Atlantic Tunas (ICCAT) through its Standing Committee on Research and Statistics (SCRS) for tunas, swordfish, billfish and some pelagic sharks. Stock size/status for Atlantic HMS assessed by ICCAT SCRS can be measured as a ratio of the current stock size (B) over the biomass at maximum sustainable yield (B_{MSY}). For internationally-assessed Atlantic HMS, a species is considered overfished if $B_{YEAR} < B_{MSY}$ (Fig 1). . Under the applicable stock determination criteria used for Atlantic HMS assessed domestically, a species is considered overfished when the current biomass (B) is less than the minimum stock size threshold ($B < B_{MSST}$) ($MSST$). B_{MSY} is also generally considered to be equal to the biomass target for these stocks. The biomass, B , can fall below B_{MSY} without causing the stock to be declared overfished as long as B remains above B_{MSST} (Fig 2).

Information on domestic and internally-assessed Atlantic HMS are available in the Atlantic HMS Stock Assessment and Fishery Evaluation (SAFE) report (see Ch 2 of the most recent SAFE available at <https://bit.ly/3YzCNa4>). Due to the different methodologies, status determination criteria, and thresholds, the Atlantic HMS SAFE summarizes the current relative biomass (or proxies) provided in individual assessments. Different metrics may be used to indicate current relative biomass, including $BYEAR/B_{MSY}$, $SSBYEAR/SSB_{MSY}$, and/or $SSFYEAR/SSF_{MSY}$. Some assessments have provided a range of current relative biomass instead of a single threshold, due to a number of reasons that vary by assessment.

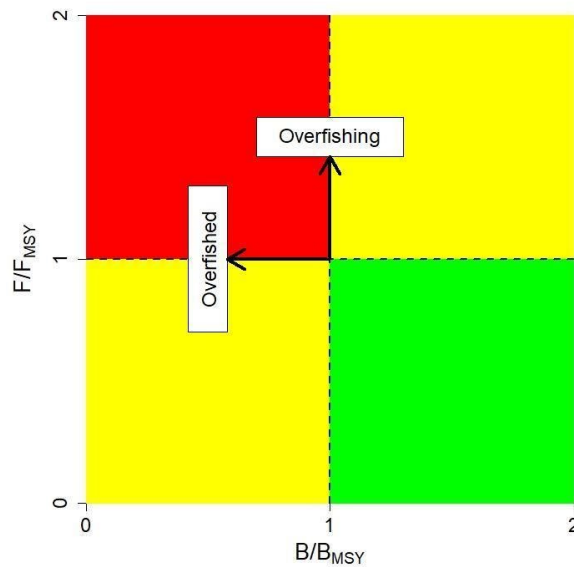


Fig 1. Illustration of the Status Determination Criteria and Rebuilding Terms for ICCAT-managed HMS Stocks

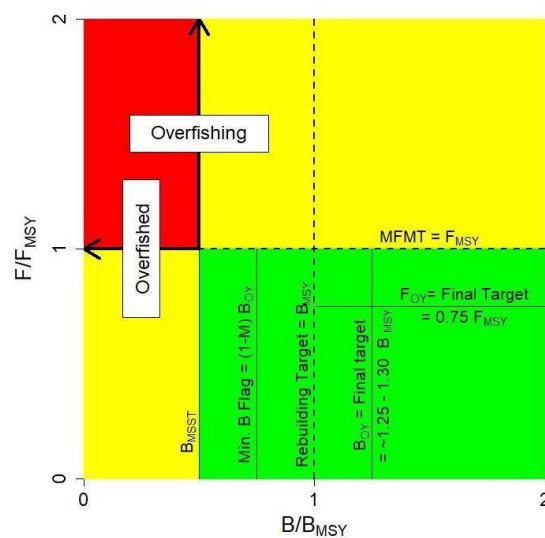


Fig 2. Illustration of the Status Determination Criteria and Rebuilding Terms for Domestically-managed HMS Stocks

Stocks that are overfished have lower overall biomass, and likely will have low genetic variation. Low genetic variation can decrease a species' ability to adapt to climate change. Large variation in reproductive success between individuals, large fluctuations in population size, and frequent local extinctions can all decrease genetic diversity (Grosberg and Cunningham 2001). Presence of these characteristics could suggest a decreased ability to adapt to changes in the environment.

How to use expert opinion: Due to its availability in the SAFE report, we recommend that you consider the current relative biomass as the basis for this sensitivity attribute. You should consider the status of the stock along with relative biomass in scoring each species. An assessment resulting in a relative biomass range that spans multiple bins or multiple reported outcomes might be interpreted to have greater uncertainty. For species with a range of relative biomass which spans multiple bins, you could consider a midpoint or your knowledge of an assessment's model or projections to tally your votes. If a direct measure of biomass is not available, biomass proxies (such as survey indices or spawning stock biomass) may be used. For data-poor stocks with an unknown status, or stocks that are analyzed as part of a species group, use your expert opinion to estimate the stock size and rate the data quality accordingly. Also, if a stock has known low genetic diversity, adjust your ranks accordingly. If nothing is available to guide your scoring, consider placing one vote in each bin, and add the fifth vote to the bin that you believe is most plausible.

Stock Size/Status Bins:

For Internationally-Assessed Species (Tunas, Swordfish, Billfish, some Pelagic Sharks):

1. **Low:** $B/B_{MSY} \geq 1.25$ (or proxy)
2. **Moderate:** For Tunas, Swordfish and Billfish: $B/B_{MSY} \geq 1.0$ but < 1.25 (or proxy)
3. **High:** For Tunas, Swordfish and Billfish: $B/B_{MSY} \geq 0.5$ but < 1.0 (or proxy)
4. **Very High:** $B/B_{MSY} < 0.5$

Stock Size/Status Bins:

For Domestically-Assessed *Sharks*:

1. **Low:** $B/B_{MSST} \geq 1.25$ (or proxy)
2. **Moderate:** $B/B_{MSST} \geq 1$ but < 1.25 (or proxy)
3. **High:** $B/B_{MSST} \geq 0.5$ but < 1 (or proxy)
4. **Very High:** $B/B_{MSST} < 0.5$

Other Stressors

Goal: To account for conditions that could increase the stress on a stock and thus decrease its ability to respond to changes.

Relationship to climate change: In most cases but not all, climate change is predicted to exacerbate the effects of other stressors. Fish stocks that are already being affected by other stressors are likely to have faster and more acute reactions to climate change.

Background: A stress is an activity that induces an adverse effect and therefore degrades the condition and viability of a natural system (Groves et al. 2000; EPA 2008). This attribute attempts to take into account interactions between climate change and other stressors already impacting fish stocks. Some examples of other stressors include: habitat degradation, invasive species, disease, pollution, and hypoxia (including vertical habitat compression, Zhang et al. 2009). Although climate change is not currently the biggest threat to many natural systems, its effects are projected to be an increasingly important source of stress in the future (Mooney et al. 2009). Consideration of observed and projected impacts of climate change in the context of other environmental stressors is essential for effective planning and management.

How to use expert opinion: For the purpose of this assessment, we are looking for detrimental impacts from other stressors. We have provided examples of other stressors that may be impacting stocks, but the list is not exhaustive. If the stock being scored is suffering from a known or suspected stressor that is not listed below, adjust the score appropriately. It is expected that in some cases, impacts of climate change could create positive impacts (e.g., reduction in predators). If you suspect positive impacts, adjust tallies toward the lower bins as appropriate. We are not including fishing pressure as a stressor here as it is covered under the “stock size/status” attribute.

Example stressors the stock may be experiencing:

- The habitat on which the stock depends is degraded. Examples include anthropogenic effects or changes to freshwater input, stratification, storm intensity, and hypoxia (including vertical habitat compression).
- The stock is currently exposed to detrimental levels of pollution (chemical and/or nutrient).
- The stock has experienced a known increase in parasites, disease, or harmful algal bloom exposure.
- The stock has experienced a detrimental impact due to a change in the food web. Examples include increases in the abundance of predators or competitors, or the introduction of an invasive species that negatively impacts the stock. Do not include changes to prey here as they are covered under the “prey specificity” attribute.

Other Stressors Bins:

1. **Low: Stock is experiencing no known stress other than fishing.** Stock is experiencing no more than one known stressor.
2. **Moderate: Stock is experiencing limited stress other than fishing.** Stock is experiencing no more than two known stressors.
3. **High: Stock is experiencing moderate stress other than fishing.** Stock is experiencing no more than three known stressors.
4. **Very High: Stock is experiencing high stress other than fishing.** Stock is experiencing four or more known stressors.

Population Growth Rate

Goal: To estimate the relative productivity of the stock.

Relationship to climate change: More productive stocks are, in general, better suited to rebound after the population is stressed by changes in the environment, such as climate change.

Background: Population growth rate is defined as the maximum population growth that would be expected to occur under natural conditions (e.g., no fishing). The amount the population changes over time can be attributed to births, deaths, emigration, or immigration of individuals between separate populations (EPA 2009). If direct measurements of population growth rate (r) are unavailable, other biological reference points that are correlated with population growth rate can be used: von Bertalanffy growth rate (k), age at maturity, maximum age and natural mortality. Scoring bins for these proxies were modified from Musick (1999) by an analysis of 141 marine fish species that were considered to be representative of U.S. fisheries (Patrick et al. 2009).

How to use expert opinion: Multiple proxies may be used to inform the final score, but the accuracy and precision of the different proxies should be considered. For example, a stock with a “good” estimate of age at maturity is in the range for a “High” score, and a “fair” estimate of maximum age is in the range for the “High” scoring bin. In that case, the scorer should use their expert opinion to weight their response according to their confidence in the estimates. If no estimates are available, estimate a relative score for the stock across a continuum of r -selected (low) vs. k -selected (high) species.

Population Growth Rate Bins:

Parameter	Low	Moderate	High	Very High
Intrinsic rate of increase (r)	> 0.50	0.16 - 0.50	0.05 - 0.15	< 0.05
von Bertalanffy K	> 0.25	0.16 - 0.25	0.11 - 0.15	≤ 0.10
Age at maturity	< 2 yrs	2 - 3 yrs	4 - 5 yrs	> 5 yrs
Maximum age	< 10 yrs	11 - 15 yrs	15 - 25 yrs	> 25 yrs
Natural mortality (M)	> 0.50	0.31 - 0.50	0.21 - 0.30	< 0.2

Mobility and Dispersal of Early Life Stages

Goal: To estimate the ability of the stock to colonize new habitats when/if their current habitat becomes less suitable.

Relationship to climate change: In general, the greater the dispersal of larvae, the better its ability to respond to climate change. Wide distribution of eggs, larvae, and young of year can lead to greater ability to colonize new habitats in areas that are suitable for survival. Conversely, if a stock has limited larval distribution and the habitat in the localized area becomes unsuitable, then the stock is more likely to be negatively affected.

Background: For marine species, extended larval and young of year dispersal is an important strategy for colonizing new areas. Duration of the larval or young of year stage may impact dispersal distance and stock persistence. Jablonski and Lutz (1983) found that marine invertebrates with relatively long planktonic larval stages were more persistent in the fossil record than those species with non- planktonic larvae and had lower extinction rates. Early life stage dispersal is affected by a number of factors including spawning, advection, diffusion, larval behavior, planktonic duration, planktonic survival, and settlement habitat (Pineda et al. 2008, Hare and Richardson in press). In general, studies have found that spawning time and place and planktonic duration are key factors, but the other factors can be important in specific situations.

How to use expert opinion: The main point of this attribute is to estimate dispersal ability. If a stock has a relatively short larval duration, but is known to disperse large distances, or if the larvae are able to influence dispersal through selective tidal stream transport, adjust your tallies accordingly. Keep in mind that long-distance dispersal of only a small fraction of the larvae could still be adequate for colonization of new areas in a changing climate. For elasmobranchs that have evolved life history strategies that produce a smaller number of well-developed offspring, the impact of this attribute will be reduced. For elasmobranchs with live birth, dispersal will occur while in utero and should be scored as low to moderate. For elasmobranchs with egg cases, egg dispersal will be more limited, but juveniles will have the ability to disperse if needed so these stocks should be scored as moderate to high. Bins were modified from Pecl et al. (2014).

Mobility and Dispersal of Early Life Stages Bins:

1. **Low: Highly dispersed eggs, larvae, or young of year.** Duration of planktonic eggs and larvae greater than 8 weeks and/or larvae or young are dispersed >100 km from spawning locations.
2. **Moderate: Moderately dispersed eggs, larvae, or young of year.** Duration of planktonic eggs and larvae less than 8 but greater than 2 weeks and/or larvae or young are dispersed 10-100 km from spawning locations.
3. **High: Low larval dispersal or young of year.** Duration of planktonic eggs and larvae less than 2 weeks and/or larvae or young typically found over the same location as parents.

- 4. Very High: Minimal larval dispersal or young of year.** Benthic eggs and larvae or little to no planktonic early life stages

Adult Mobility

Goal: To estimate the ability of the stock to move to a new location if their current location changes and is no longer favorable for growth and/or survival.

Relationship to climate change: Site-dependent species that are unable to move to better habitat when a location becomes unfavorable are less able to adapt to environmental change than highly mobile species.

Background: As climate change occurs, habitats that were once suitable may change and no longer be able to sustain a given stock of fish. Similarly, what was once an unsuitable habitat may become suitable. A stock can survive changes in habitat as long as they have the ability to disperse from unsuitable habitat and find new, suitable habitat. This can occur through larval dispersal and settlement (covered under the “Dispersal of Early Life Stages” attribute) or through adult mobility. Species can be limited in their mobility by physical or behavioral (e.g., won’t swim across open ocean) barriers.

How to use expert opinion: This attribute represents a continuum from sessile to highly migratory organisms. Use your expert opinion to place the stock in question in the appropriate bin according to its physical and behavioral ability to move.

Adult Mobility Bins:

1. **Low: Non-site dependent.** The stock is highly mobile and non-site dependent.
2. **Moderate: Site dependent but highly mobile.** The stock has site-dependent adults capable of moving from one site to another if necessary.
3. **High: Site dependent with limited mobility.** The stock has site-dependent adults that are restricted in their movement by environmental or behavioral barriers.
4. **Very High: Non-mobile.** The stock has sessile adults.

Reproductive Cycle

Goal: To determine if the duration of the reproductive cycle for the stock could limit the ability of the stock to successfully reproduce if necessary conditions are disrupted by climate change.

Relationship to climate change: It is assumed that stocks that reproduce over an extended period of time will be more likely to be successful in a changing environment. Conversely, stocks that reproduce all at once in major events are more likely to experience recruitment failure with potential changes in environmental conditions.

Background: Reproductive characteristics describe the reproductive activity of a stock (in aggregate, not individually) over a particular time frame. If a stock reproduces several times per year across a variety of seasons, then they will likely be less susceptible to climate change because their reproductive events are not dependent on just one set of very specific conditions (e.g., phenological events). Increased reproductive events, a type of bet hedging, also help to protect against vulnerabilities associated with single reproductive aggregations (see the “Complexity in Reproductive Strategy” attribute). Similarly, stocks that reproduce seasonally are also less likely to adapt to climate change as they are dependent on environmental conditions historically present during a given season that may not persist through time. For example, spring-like conditions and related activities have occurred progressively earlier since the 1960s (Walther et al. 2002) and changes in reproductive season and location have already been observed and predicted to continue (Shoji et al. 2011; Rijnsdorp et al. 2009). Note: We are describing the reproductive activity of the entire stock, not the individual. In other words, we are interested in the time from when reproduction commences until when it ends, not how long a single individual reproduces.

How to use expert opinion: It is impossible to distill every potential reproductive cycle into 4 scoring bins. The below bins are rough breaks in a continuum of possibilities. If a species does not fit the below bins, use your expert judgment to best score the species based on the above discussion. For stocks (such as elasmobranchs) that are born as fully developed juveniles capable of long distance movements, there is less concern over a short hatching/mating period, and these stocks should be ranked low to moderate.

Reproductive Cycle Bins:

1. **Low: Consistent throughout the year.** Stocks that reproduce continuously throughout the year without a defined “reproductive season” are considered to be at the lowest risk of suffering from adverse effects of climate change. Example: a stock that reproduces daily or monthly.
2. **Moderate: Several spawning events throughout the year.** Stocks that reproduce several times per year and reproduce across more than one season have a moderate risk. Example: a stock that reproduces in both the spring and summer.
3. **High: Several spawning events per year within a confined time frame.** Stocks that may reproduce several times per year but all reproductive events in that year take place in one season have a high risk of being affected by climate change. Example: the

reproductive season occurs once a year and lasts over a period of less than 3 months.

4. **Very High: One spawning event per year.** Stocks that require very specific environmental/social queues to initiate reproduction and that only reproduce once per year are at the highest risk level for being affected by climate change. Example: the spawning season occurs once a year over a brief period of time.

Site Fidelity

Goal: To assess the degree to which individuals utilize the same sites year after year.

Relationship to climate change: If a site that individuals return to is impacted by climate change, those individuals are expected to be impacted as well (Laidre et al. 2008). Stocks that exhibit weak site fidelity may be better suited to adapt to changing climate conditions and increased climate variability than stocks with strong site fidelity (Abrahms et al. 2018). Stocks with strong site fidelity are more likely to require shifts in distribution beyond their traditional sites to adapt to climate-driven changes that impact those sites.

Background: Site fidelity is defined as the tendency to remain in, or return to, the same site year after year (Switzer 1993). Individuals that remain in or return to the same locations (e.g., foraging grounds, haul-out sites) display high site fidelity.

Here we assess site fidelity as the precision to which individuals remain in or return to the same geographic areas year after year. Individuals that remain in or return to a smaller, or more precise, area (such as specific beaches, islands, or bays) exhibit a higher degree of site fidelity.

Those individuals that remain in or return to the same general region with less precision exhibit a lower degree of site fidelity. Beyond the degree of site fidelity, the number of sites that the individuals of a stock show fidelity toward has an impact on the sensitivity of the stock to climate change. Stocks that show a high degree of site fidelity to a single, same site are highly sensitive to climate change. As the proportion of individuals within a stock exhibiting high site fidelity to the same single site increases, the sensitivity to climate change also increases. Stocks that show a high degree of site fidelity, but the individuals of the stock show fidelity to different locations from the other individuals, are also sensitive to climate change, but to a lesser degree than stocks with fidelity to one or two locations. A greater number of unique sites that the stock shows fidelity toward reduces the impact of high site fidelity on sensitivity to climate change.

How to use expert opinion: This attribute represents a continuum from organisms that display no site fidelity to strong site fidelity throughout its adult life stage. Remaining in or returning to specific conditions (e.g., eddies, fronts) that shift in space fits better with the habitat specificity attribute than with site fidelity. Use your expert opinion to place the stock in question in the appropriate bin according to its behavior. Homing behavior for spawning should not be considered here as it is accounted for in the “Reproductive strategy sensitivity” attribute. For this attribute, we define site-dependent stocks as those whose adults are site-attached (i.e. spend their entire adult phase in one limited location) or routinely return to the same site, whether for foraging, shelter, overwintering, or another purpose excluding reproduction.

Site Fidelity Bins:

1. Low: Individuals display no site fidelity.
2. Moderate: Individuals display a low degree of site fidelity (i.e., archipelagos or coastlines of a general region, maintaining presence along specific ocean currents or gyres). The species has several highly specific locations of site fidelity.

3. High: Individuals display a moderate degree of site fidelity (i.e., specific bays or estuaries) for foraging, shelter, overwintering, or other needs. The species has few highly specific locations of site fidelity.
4. Very High: Individuals display a high degree of site fidelity (i.e., specific bays, estuaries, or inlets) for both foraging, shelter, overwintering, or other needs. Individuals remain permanently in the areas they display site fidelity, and there are few to one specific locations of site fidelity.

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